PROJECT PRESENTATION

No More Traffic!

TITLE : NO MORE TRAFFIC

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PROBLEM STATEMENT

What the problem is

- Design a traffic control logic for a “Stop-Free Corridor” in a smart city.
- The city road network is modeled as a directed graph, where intersections are nodes and roads are edges with fixed travel time.
- Vehicles move at a uniform speed, so the arrival time at each intersection is predetermined.
- Traffic signals must be scheduled so that each vehicle reaches the intersection exactly when the signal turns green.
- At intersections with crossing paths, both directions cannot be green at the same time.

PROBLEM STATEMENT

Why the problem needs to be solved

- To allow groups of vehicles (platoons) to travel through multiple intersections without stopping.
- To synchronize traffic signals across the network using precise green-light timing.
- To maximize the number of vehicles that complete their journey smoothly without being blocked by conflicting traffic signals.
- To improve traffic efficiency and smart city traffic management.

PROJECT OBJECTIVES

01.

Traffic Network Modeling
Model the city traffic system as a directed graph, where intersections are nodes and roads are edges with fixed travel time.

02.

Vehicle Arrival Time Calculation
Compute the exact arrival time of vehicles at each intersection based on travel time and entry time.

03.

Green Signal Synchronization
Schedule green-light windows so vehicles reach intersections exactly when the signal is green.

04.

Conflict-Free Scheduling
Ensure crossing paths are not green simultaneously and maximize the number of vehicles that travel without stopping.

ALGORITHMS

01. **Graph-Based Scheduling Algorithm** – The traffic network is represented as a directed graph where intersections are nodes and roads are edges.

02. **Greedy Scheduling Approach** – Green signal windows are assigned based on vehicle arrival times to allow smooth movement.

03. **Conflict Checking Mechanism** – Ensures that intersecting paths are not given green signals at the same time.

04. **Optimization Strategy** – The schedule is adjusted to maximize the number of vehicles that pass through the network without stopping.

PROPOSAL ALGORITHM

ALGORITHM NAME:
GRAPH-BASED TRAFFIC SIGNAL SCHEDULING ALGORITHM

01

Network Representation
Represent the city traffic system as a directed graph where intersections are nodes and roads are edges.

02

Input Vehicle Paths
Take the vehicle paths, entry time, and travel time between intersections as input.

03

Calculate Arrival Time
Compute the arrival time of vehicles at each intersection based on travel time.

04

Green Signal Assignment
Assign green signal windows at intersections according to the vehicle arrival times.

PROPOSAL ALGORITHM

05

Conflict Detection
Check if intersecting paths receive green signals at the same time.

06

Schedule Optimization
Adjust signal timings to avoid conflicts and maximize vehicles passing without stopping.

07

Vehicle Flow Validation
Verify that vehicles can pass through intersections without stopping based on the schedule.

08

Final Signal Schedule Output
Generate the final green-light schedule for all intersections in the network.

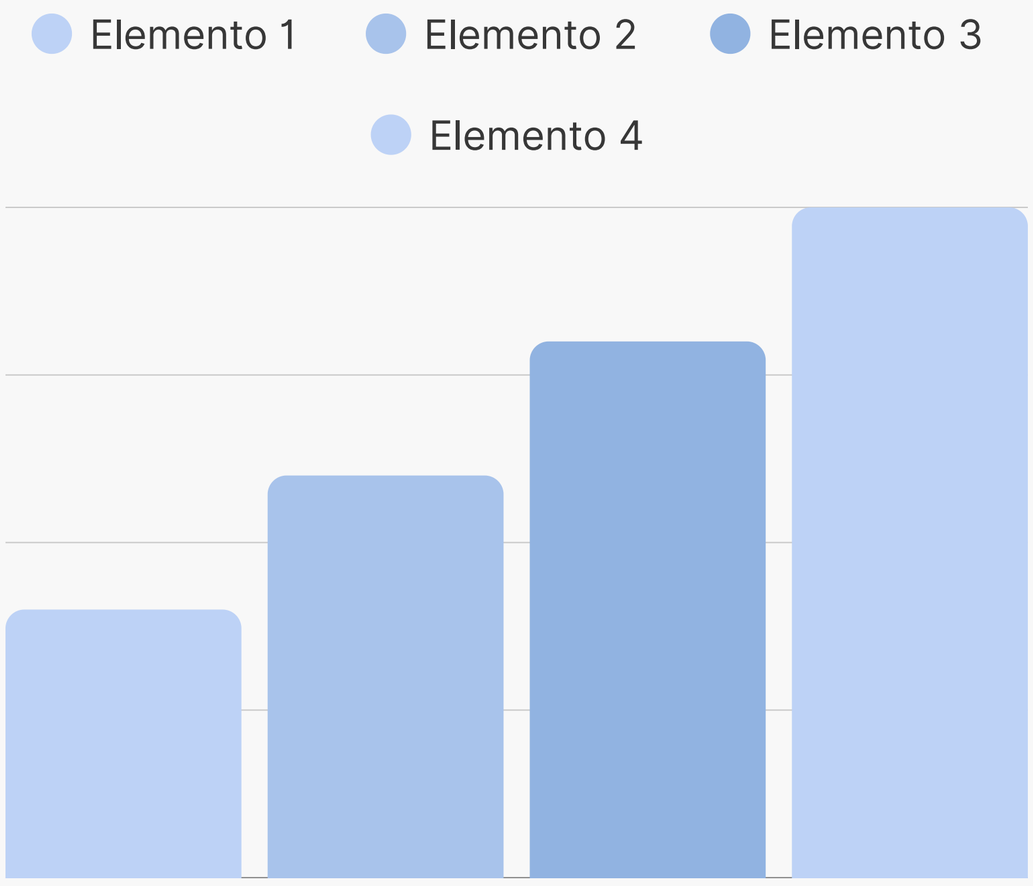
PSEUDOCODE

```
Algorithm StopFreeTrafficScheduling

Input:
   Graph G(V, E)
   Travel time W_e for each edge
   Set of vehicle paths P
Output:
   Green signal schedule for each intersection
1. For each path p in P
2.     current_time ← entry_time(p)
3.     For each edge e in path p
4.         current_time ← current_time + travel_time(e)
5.         arrival_time ← current_time
6.         Schedule green signal at intersection for
arrival_time
7.         If conflict occurs with another path
8.             Adjust signal timing
9.         End If
10.    End For
11. End For

12. Return final green signal schedule
```

TIME AND SPACE COMPLEXITY

Time Complexity

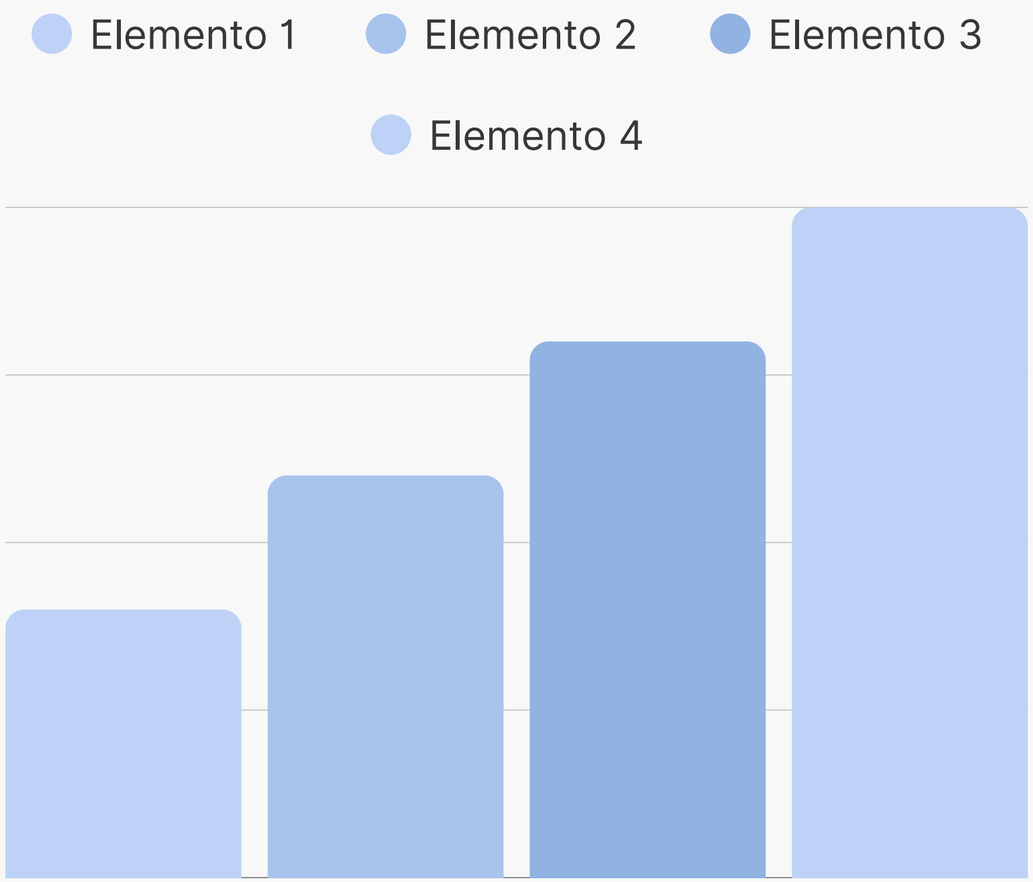
- Let V be the number of intersections and E be the number of roads.
- The algorithm processes each vehicle path and calculates arrival times.
- Signal scheduling and conflict checking are done at intersections.
- Overall Time Complexity: O(P × L)
  - P = number of vehicle paths
  - L = average intersections per path



TIME AND SPACE COMPLEXITY

Space Complexity

- Space is required to store the graph representation, vehicle paths, and signal schedules.
- Graph storage requires O(V + E) space.
- Additional memory is used for arrival times and schedules.
- Overall Space Complexity: O(V + E + P)



IMPLEMENTATION

Code Snippet

```
#include <stdio.h>

int calculateArrivalTime(int entryTime, int
travelTime) {
    return entryTime + travelTime;
}
int main() {
    int entryTime = 5;
    int travelTime = 3;

    int arrivalTime =
calculateArrivalTime(entryTime, travelTime);

    printf("Arrival Time at Intersection: %d\n",
arrivalTime);

    return 0;
}
```

01

Programming Language

The algorithm is implemented using **C Programming Language** because it provides efficient memory management and fast execution for algorithm-based applications.

02

Data Structures Used

The traffic network is represented using **arrays and graph** structures.
Intersections are stored as nodes and roads are stored as edges with travel time.

03

Implementation Logic

The program calculates vehicle **arrival times at intersections**, schedules green signals, and checks for conflicts between intersecting paths to allow smooth traffic flow.

RESULTS AND ANALYSIS

INPUT

```
Enter number of intersections: 4
Enter number of roads: 4
Enter roads (u v travel_time):
0 1 3
1 2 4
2 3 2
1 3 5
Enter number of platoons: 2

Platoon 1
Entry time: 2
Path length: 4
Enter path: 0 1 2 3

Platoon 2
Entry time: 3
Path length: 3
Enter path: 0 1 3
```

RESULTS AND ANALYSIS

OUTPUT

```
Platoon 1 Simulation
Time 5 : Intersection 1 GREEN for 0 -> 1
Time 9 : Intersection 2 GREEN for 1 -> 2
Time 11 : Intersection 3 GREEN for 2 -> 3

Platoon 2 Simulation
Time 6 : Intersection 1 GREEN for 0 -> 1
Time 11 : Intersection 3 RED


------ RESULTS ------
Successful Platoons: 1
Blocked Platoons: 1
Success Rate: 50.00%

Traffic Load per Intersection
Intersection 0 : 0 vehicles
Intersection 1 : 2 vehicles
Intersection 2 : 1 vehicles
Intersection 3 : 2 vehicles

--------------------------------
Process exited after 179.3 seconds with return value 0
Press any key to continue . . .
```

CONCLUSION

01

Key Outcome
The proposed Graph-Based Traffic Signal Scheduling Algorithm successfully synchronizes traffic signals across multiple intersections.

02

System Benefit
The algorithm calculates vehicle arrival times and schedules green signals, allowing vehicles to move smoothly without stopping.

03

Overall Impact
This approach helps reduce traffic congestion, minimize waiting time, and improve traffic flow, supporting efficient smart city traffic management.

THANK YOU VERY MUCH!

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